THE COMPOSITE OF TWO MAPS OF ADMISSIBLE SEQUENCES IS HOMOTOPIC TO ZERO

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-nilpotent (source lines 840-867); the proof environment begins two lines below it and is its own proof, closing at source line 876.

Editorial note. The proof invokes the author's lemma lemma-make-commute-map by label; that lemma's statement and its own proof are retained as a context span placed before the counted statement, so the reference resolves inside the assembled package and no source pointer is declared.

Editorial note. The statement contains two Xy-pic squares displayed inside vcenter; both are part of the retained source range and are reproduced as written, so no figure file is needed.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-dga/preamble.tex, 217 lines), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Differential Graded Algebra chapter, counted once: the composite of two maps of admissible short exact sequences that commute up to homotopy is homotopic to zero.

Allowed standard prerequisites (editorial).

- Admissible short exact sequence of dg modules. A short exact sequence of dg modules splitting as graded modules, with the associated maps between the outer and middle terms.. The three sequences provide the maps whose images and kernels are compared (source lines 843-844, 873-874).
- Commuting up to homotopy and strict commutation. Two squares of dg module maps commute up to homotopy; a map homotopic to a strictly commuting one can be substituted.. The hypothesis says the two squares commute up to homotopy and the proof replaces the maps by homotopic ones making the relevant squares commute strictly (source lines 845-865, 870-872).
- Replacement lemma. If a square commutes up to homotopy and one of its horizontal maps is an admissible monomorphism, then the vertical map can be replaced by a homotopic one making the square commute (the author's lemma lemma-make-commute-map, retained as context).. It is invoked to obtain the strict squares, first for b and then for b' (source line 870).

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Lemma 0.1. *Let (A, d) be a differential graded algebra. Let*

$$\begin{array}{ccc} K & \xrightarrow{f} & L \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ M & \xrightarrow{g} & N \end{array}$$

be a diagram of homomorphisms of differential graded A-modules commuting up to homotopy.

(1) *If f is an admissible monomorphism, then b is homotopic to a homomorphism which makes the diagram commute.*
(2) *If g is an admissible epimorphism, then a is homotopic to a morphism which makes the diagram commute.*

Proof. Let $h : K \to N$ be a homotopy between bf and ga, i.e., $bf - ga = \mathrm{d}h + h\mathrm{d}$. Suppose that $\pi : L \to K$ is a graded A-module map left inverse to f. Take $b' = b - \mathrm{d}h\pi - h\pi\mathrm{d}$. Suppose $s : N \to M$ is a graded A-module map right inverse to g. Take $a' = a + \mathrm{d}sh + sh\mathrm{d}$. Computations omitted. □

Lemma 0.2. *Let (A, d) be a differential graded algebra. Let $0 \to K_i \to L_i \to M_i \to 0$, $i = 1, 2, 3$ be admissible short exact sequence of differential graded A-modules. Let $b : L_1 \to L_2$ and $b' : L_2 \to L_3$ be homomorphisms of differential graded modules such that*

$$\begin{array}{ccccc} K_1 & \longrightarrow & L_1 & \longrightarrow & M_1 \\ {\scriptstyle 0}\downarrow & & {\scriptstyle b}\downarrow & & {\scriptstyle 0}\downarrow \\ K_2 & \longrightarrow & L_2 & \longrightarrow & M_2 \end{array} \quad \text{and} \quad \begin{array}{ccccc} K_2 & \longrightarrow & L_2 & \longrightarrow & M_2 \\ {\scriptstyle 0}\downarrow & & {\scriptstyle b'}\downarrow & & {\scriptstyle 0}\downarrow \\ K_3 & \longrightarrow & L_3 & \longrightarrow & M_3 \end{array}$$

commute up to homotopy. Then $b' \circ b$ is homotopic to 0.

Proof. By Lemma 0.1 we can replace b and b' by homotopic maps such that the right square of the left diagram commutes and the left square of the right diagram commutes. In other words, we have $\mathrm{Im}(b) \subset \mathrm{Im}(K_2 \to L_2)$ and $\mathrm{Ker}(b') \supset \mathrm{Im}(K_2 \to L_2)$. Then $b' \circ b = 0$ as a map of modules. □